

Unit 1: Data Representation And Number System	1	What is a number system? Explain decimal, binary, octal and hexadecimal number systems.
	2	Convert (110101)₂ into decimal and octal.
	3	Convert (9F)₁₆ into binary and decimal.
	4	Explain binary addition with a suitable example.
	5	Perform binary subtraction using 1's complement method.
	6	Explain binary subtraction using 2's complement method with example.
	7	Explain binary multiplication and division (simple method).
Unit 2: Introduction to The Computer And Computer Peripherals	8	Explain different generations of computers.
	9	Draw and explain the block diagram of a computer system.
	10	Explain different types of input devices with examples.
	11	Explain output devices and their applications.
	12	Differentiate between primary and secondary storage devices.
	13	Compare micro computer, mini computer and super computer.
Unit 3: Introduction And Installation of Operating System	14	What is an operating system? Explain its functions.
	15	Explain different types of operating systems.
	16	Describe hardware and software requirements for OS installation.
	17	Explain step-by-step installation of an operating system.
	18	Explain the computer booting process.
	19	Explain the need and installation of device drivers.
Unit 4: Introduction to Communication Device	20	Explain USB, HDMI and VGA ports.
	21	Explain Serial and Parallel ports.
	22	What are OMR, MICR and OCR ? Explain their uses.
	23	Explain MODEM and NIC with diagram.
	24	Differentiate between data and information.
	25	Explain programmed I/O , interrupt and DMA .
Unit 5: Introduction to Generation Languages And Processor	26	Define language processor. Explain compiler, interpreter and assembler.
	27	Explain machine language and assembly language.
	28	Differentiate between low-level and high-level languages.
	29	What are scripting languages? Give examples.
	30	Write a short note on next-generation languages.
	31	Explain the role of web technology in modern applications.